

INCREASE REVENUE AND PROFITABILITY OF A LAUNDRY IRONING BUSINESS AND PLAN FOR FUTURE EXPANSION

BDM CAPSTONE PROJECT

FINAL Submission

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1. **Executive Summary**

M/s. Suresh Laundry Ironing Shop, located on New Street, Aminjikarai, Chennai, has been serving the local community since 2000. Despite the dynamic changes in the surrounding residential and commercial areas, the shop's business model has largely remained the same over the past two and a half decades. The objective of this project is to improve revenue and profitability and outline a strategy for business expansion.

The project began with a detailed study of the shop's operations, environment, and financials. Data collection methods included questionnaires, interviews, behavioural observation, and time series analysis over a 30-day period. The collected data were normalized and analysed using both quantitative and qualitative techniques. Analytical methods such as volume and revenue analysis, mean/median/mode calculations, return ratios, fixed and variable cost analysis, SWOT analysis, and process re-engineering were employed. These methods provided insights into operational inefficiencies and potential areas for improvement.

Based on the findings, several proposals were developed. Key suggestions include replacing coal-powered irons with gas-powered ones to reduce costs and environmental impact and regularizing laundry collection and distribution schedules to optimize workload distribution throughout the week. Additionally, proposals were made to increase revenue by expanding into services like laundry washing and dry cleaning, which offer higher returns per minute of work. These improvements aim to increase both customer satisfaction and the shop's overall profitability.

The data-driven proposals presented in this report are expected to significantly enhance the shop's efficiency, revenue, and potential for future business growth.

2. Detailed Explanation of Analysis Process/Method

The Data from 29th July 2024 to 27th August 2024, a 30-day period, was collected from the shop. The below methodologies were used for data collection:-

- a) **Questionnaires and Interviews:** Prepared and conducted with the shop owner to understand the business processes, cost structures, and challenges in detail.
- b) **On-Site Observations:** Regular visits to the shop during various times of the day to collect real-time data, assess workload fluctuations, and understand customer interaction.
- c) **Log book for Daily Recording:** A log book was provided to the shop owner, enabling the recording of daily transactions and service details during periods when I was not present.
- d) **Environmental Study:** A thorough observation of the business environment was conducted, considering customer behavior, peak hours and other external factors impacting operations.
- e) **Market Research:** This involved analyzing local competitors and studying expansion opportunities, helping to propose strategies for business growth.

The data collected daily from the above techniques was placed in an excel sheet to be used for analysis. The data collected was normalized taking into account missing data, outliers and consistency to facilitate Quantitative and Qualitative analysis.

2.1. Qualitative Analysis

2.1.1. Questionnaire

(Refer Questionnaire – [LINK](#))

In order to understand the operation of the shop a set of questions were prepared and had the questionnaire run past the owner. Along with this informal discussions were held with the owner, which also help build a good rapport paving the way for gathering more details. This provided valuable insight on the fundamentals, complexities and operational processes involved in running the business. This helped in understanding the chain of activities involved, time taken to perform each activity, identifying redundancy / inefficiency and help to re-engineer the processes involved.

2.1.2. SWOT Analysis

(Refer to Figure-1 S.W.O.T.- [LINK](#))

In order to analyze the landscape business operates:-

- The **Strength (S)** of the business which it could leverage upon,
- The **Weakness (W)** of the business which it should overcome and avoid,
- The **Opportunities (O)** that the shop could explore to expand its business,
- The **Threat (T)** that should keep a close eye-on to ensure it does not negatively impact the business.

Strength Weakness Opportunity Threat (S.W.O.T.) Analysis

(S) <u>Strength</u>	• Good word of mouth reputation and Trust built in the locality over the years • Monopoly in the locality - as there are no other Laundry Ironing shops in the street • Surrounded by high-end Residential area - who have more spending power • Situated near a 4-Star Hotel, Hostels and Lodges - a potential source of business • Expertise in running the business for more than last 2-decades
(W) <u>Weakness</u>	• Lack of structured Information and Guidance available in running the business • Not enough Capital available to expand business • Peaks and Troughs for service demand challenges regular income generation • Service downtime with Coal Powered Iron box • Limited services offered - not able to take up new business orders
(O) <u>Opportunity</u>	• Situated near a 4-Star Hotel, Hostels and Lodges - a potential source of business • Leverage the good reputation and monopoly enjoyed in the location, to increase revenue and to provide new services • Distribute laundry volume evenly to reduce peak work stress and to enable taking up adhoc new business orders • Process re-engineer to bring in efficiency, increase volume of laundry serviced and improve revenue/profit margin • Reduce downtime by using Gas Powered Iron box • Provide own Dry Cleaning service which can be more profitable
(T) <u>Threat</u>	• Unable to service customers during peak periods, could lead to business lost permanently • High possibility of loosing monopoly if a new laundry ironing shop opens in the locality • Unable to provide new services could result in customers opting for new service providers • Using Coal Powered Iron-box is hazardous to health, as it can cause Tuberculosis

Figure–1 S.W.O.T. Analysis

2.1.3. AS-IS and TO-BE Analysis

Processes involved in running the shop was grouped together. This AS-IS process areas was further analyzed to understand activities, interdependencies and time involved at each stage.

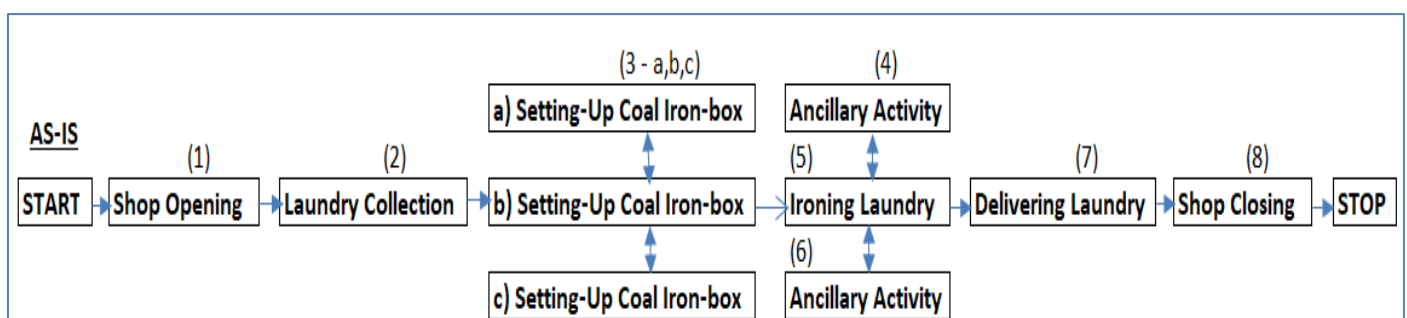


Figure-2 AS-IS and TO-BE Analysis

Restructuring by reducing/eliminating redundant process like moving away from coal powered iron-box to gas powered iron-box, streamline existing processes like adhoc collection and categorization of laundry, helped to increase efficiency. This study was extended by interpolating time taken for each process and reorganizing the routine, to unlock potential revenue growth that can be achieved.

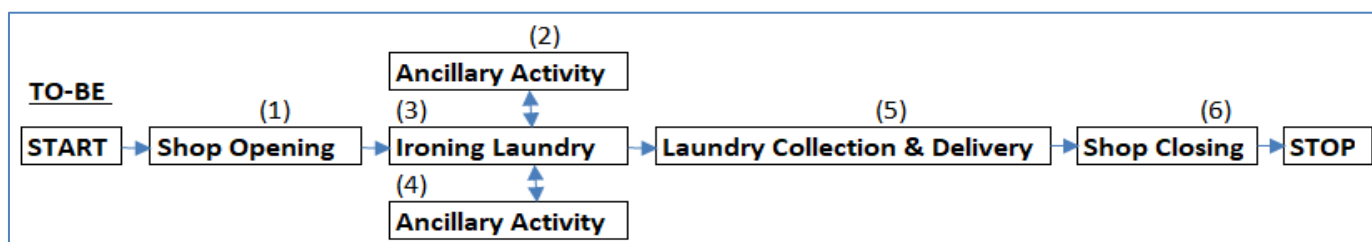


Figure-3 TO-BE Process WorkFlow

As part of Market Research, a door-to-door survey covering the residential and commercial establishments was conducted. This provided insights on customer expectations, opportunities that can be explored, current trends, etc. The survey helped understand the reason for Peaks and Troughs in volume /revenue, laundry service demand, behavioural pattern of Customers, potential for new services and help rollout plan for improving customer loyalty to regularize revenue.

2.2 Quantitative Analysis

2.2.1. Time Series Data Analysis

(Refer to Figure-4 Volume Distribution– [LINK](#))

			Total			Shirt Cotton			Shirt Nylon/Synthetic			Pant Cotton / Jeans			Pant Nylon/Synthetic			T-Shirt			Saree Cotton			Saree Silk			Churidar Salwar Kameez			Bed sheets			School Uniform Boy/Girl			
S. No	Date	Weekday	Volu me	Time	Revenue	Volu me	Time	Revenue	Volu me	Time	Revenue	Volu me	Time	Revenue	Volu me	Time	Revenue	Volu me	Time	Revenue	Volu me	Time	Revenue	Volu me	Time	Revenue	Volu me	Time	Revenue	Volu me	Time	Revenue				
1	29-07-24	Monday	118	579	1,670	20	60	200	8	16	80	20	100	200	10	40	100	16	32	160	7	35	140	4	32	120	32	256	640	1	8	30	0	0	0	
2	30-07-24	Tuesday	125	624	1,800	20	60	200	6	12	60	18	90	180	6	24	60	22	44	220	10	50	200	2	16	60	41	328	820	0	0	0	0	0		
3	31-07-24	Wednesday	117	576	1,670	10	30	100	15	30	150	24	120	240	10	40	100	12	24	120	8	40	160	2	16	60	30	240	600	2	16	60	4	20	80	
4	01-08-24	Thursday	126	612	1,760	16	48	160	16	32	160	22	110	220	10	40	100	14	28	140	9	45	180	1	8	30	36	288	720	1	8	30	1	5	20	
5	02-08-24	Friday	119	554	1,650	8	24	80	20	40	200	18	90	180	16	64	160	15	30	150	10	50	200	4	32	120	28	224	560	0	0	0	0	0		
6	03-08-24	Saturday	171	819	2,410	26	78	260	10	20	100	25	125	250	20	80	200	26	52	260	11	55	220	5	40	150	42	336	840	1	8	30	5	25	100	
7	04-08-24	Sunday	156	773	2,230	18	54	180	12	24	120	28	140	280	10	40	100	24	48	240	10	50	200	3	24	90	46	368	920	0	0	0	5	25	100	
8	05-08-24	Monday	116	566	1,640	10	30	100	14	28	140	24	120	240	6	24	60	18	36	180	8	40	160	1	8	30	32	256	640	3	24	90	0	0	0	
9	06-08-24	Tuesday	104	441	1,400	8	24	80	30	60	300	10	50	100	6	24	60	17	34	170	5	25	100	2	16	60	25	200	500	1	8	30	0	0	0	
10	07-08-24	Wednesday	98	492	1,390	8	24	80	8	16	80	15	75	150	15	60	150	11	22	110	7	35	140	0	0	0	30	240	600	0	0	0	4	20	80	
11	08-08-24	Thursday	114	549	1,580	17	51	170	14	28	140	10	50	100	14	56	140	15	30	150	5	25	100	0	0	0	38	304	760	0	0	0	1	5	20	
12	09-08-24	Friday	99	456	1,350	12	36	120	16	32	160	18	90	180	6	24	60	14	28	140	6	30	120	2	16	60	24	192	480	1	8	30	0	0	0	
13	10-08-24	Saturday	159	753	2,240	14	42	140	20	40	200	20	100	200	23	92	230	22	44	220	10	50	200	4	32	120	40	320	800	1	8	30	5	25	100	
14	11-08-24	Sunday	159	807	2,360	20	60	200	10	20	100	26	130	260	14	56	140	20	40	200	12	60	240	6	48	180	44	352	880	2	16	60	5	25	100	
15	12-08-24	Monday	101	480	1,440	10	30	100	15	30	150	12	60	120	8	32	80	16	32	160	8	40	160	2	16	60	29	232	580	1	8	30	0	0	0	
16	13-08-24	Tuesday	101	493	1,370	8	24	80	10	20	100	22	110	220	14	56	140	12	24	120	7	35	140	1	8	30	27	216	540	0	0	0	0	0	0	
17	14-08-24	Wednesday	101	493	1,500	7	21	70	12	24	120	8	40	80	10	40	100	19	38	190	6	30	120	4	32	120	31	248	620	0	0	0	4	20	80	
18	15-08-24	Thursday	105	486	1,400	20	60	200	15	30	150	16	80	160	8	32	80	11	22	110	5	25	100	0	0	0	29	232	580	0	0	0	1	5	20	
19	16-08-24	Friday	111	477	1,450	14	42	140	20	40	200	17	85	170	11	44	110	17	34	170	8	40	160	1	8	30	22	176	440	1	8	30	0	0	0	
20	17-08-24	Saturday	152	733	2,160	10	30	100	20	40	200	22	110	220	16	64	160	23	46	230	10	50	200	2	16	60	43	344	860	1	8	30	5	25	100	
21	18-08-24	Sunday	155	745	2,210	16	48	160	20	40	200	24	120	240	16	64	160	19	38	190	10	50	200	5	40	150	39	312	780	1	8	30	5	25	100	
22	19-08-24	Monday	97	428	1,290	15	45	150	14	28	140	15	75	150	9	36	90	14	28	140	8	40	160	0	0	0	20	160	400	2	16	60	0	0	0	
23	20-08-24	Tuesday	106	507	1,460	21	63	210	10	20	100	13	65	130	11	44	110	13	26	130	5	25	100	2	16	60	31	248	620	0	0	0	0	0	0	
24	21-08-24	Wednesday	82	423	1,200	4	12	40	10	20	100	16	80	160	6	24	60	8	16	80	7	35	140	0	0	0	27	216	540	0	0	0	4	20	80	
25	22-08-24	Thursday	89	442	1,240	8	24	80	12	24	120	16	80	160	8	32	80	10	20	100	5	25	100	0	0	0	29	232	580	0	0	0	1	5	20	
26	23-08-24	Friday	121	597	1,720	8	24	80	20	40	200	18	90	180	10	40	100	16	32	160	7	35	140	2	16	60	40	320	800	0	0	0	0	0	0	
27	24-08-24	Saturday	157	739	2,190	20	60	200	15	30	150	20	100	200	18	72	180	25	50	250	10	50	200	3	24	90	41	328	820	0	0	0	5	25	100	
28	25-08-24	Sunday	153	706	2,100	18	54	180	22	44	220	21	105	210	17	68	170	20	40	200	10	50	200	1	8	30	38	304	760	1	8	30	5	25	100	
29	26-08-24	Monday	84	413	1,210	10	30	100	7	14	70	14	70	140	8	32	80	11	22	110	9	45	180	1	8	30	22	176	440	2	16	60	0	0	0	
30	27-08-24	Tuesday	104	519	1,470	10	30	100	10	20	100	16	80	160	14	56	140	13	26	130	7	35	140	1	8	30	32	256	640	1	8	30	0	0	0	
Total			3,600	17,282	50,560																															
Avg (Daily)			120	576	1,685	14	41	135	14	29	144	18	91	183	12	47	117	16	33	164	8	40	160	2	16	61	33	263	659	1	6	23	2	10	40	

Figure – 4 :- Data of Volume Distribution

The main data I relied on for analysis was **Time Series Data**, where data for 30-days was collected and presented in an excel sheet. This includes the Cost of running the shop, Types and Volume of laundry serviced, Revenue generation and Time to service each laundry.

Data Normalization was performed, by reviewing and organizing them taking into account missing data, outliers and consistency.

2.2.2 Statistical Data Analysis (Refer to Figure-5 Statistical Data Chart– [LINK](#))

Applying descriptive statistical techniques like average laundry volume, time spent to service each laundry, overheads involved, etc. helped me gain an insight on the operating nature of the shop. Graphical representation of data helped in providing valuable insights on distribution of laundry during the week:-

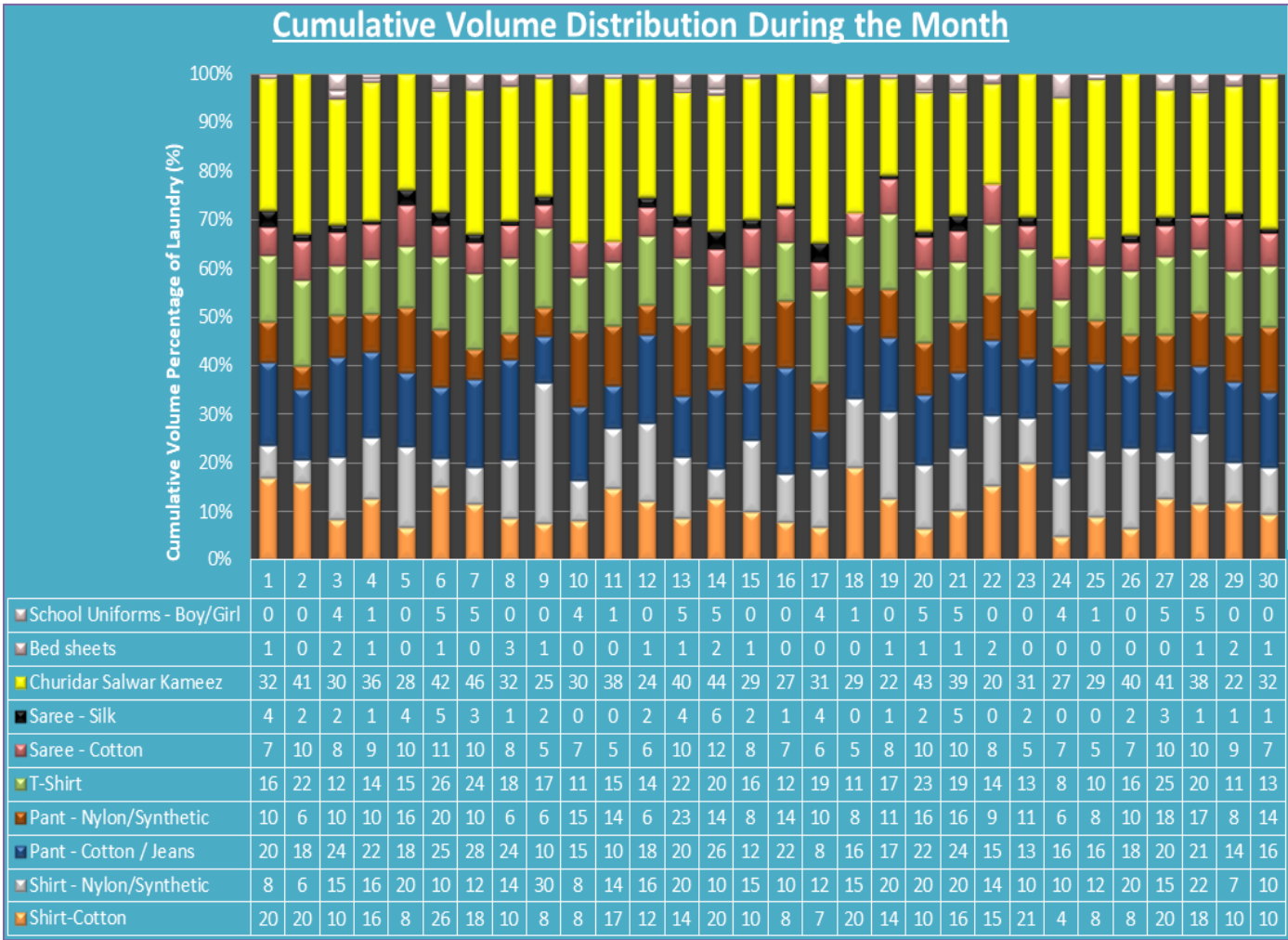


Figure – 5 :- Graphical Representation of Volume Distribution During Month

To understand the demand distribution during the week, I used a line chart to plot the volume during the week, for the entire month of August. This gave me insights on peaks and troughs during the week. (This is explained in the next section).

Bar charts were used to understand the revenue earned by the shop on the time spent per minute for each type of laundry. This gave insights on the most-profitable / high-revenue generating laundry types.

S.No.	Type of Cloth	RPM (Rate/Time)	Volume
1	Dry Cleaning - Silk Saree (Outsourced)	10.00	15
2	Washing & Ironing - Saree	5.33	10
3	Shirt - Nylon/Synthetic	5.00	431
4	Washing & Ironing - Pant/Shirt Set	4.33	40
5	T-Shirt	4.00	493
6	Saree - Cotton	4.00	240
7	1-Set of School Uniform - Boy/Girl	4.00	60
8	Saree - Silk	3.75	61
9	Bed sheets	3.75	23
10	Shirt - Cotton	3.33	406
11	Pant - Nylon/Synthetic	2.50	350
12	1-Set Churidar Salwar Kameez	2.50	988
13	Pant - Cotton / Jeans	2.00	548

(* Rank based on RPM)

Figure – 6 :- Return Per Minute (RPM)

Statistics like Mean, Median, Mode, Minimum and Maximum analysis on data collected helped provide useful information on the work load the shop experiences during different time of the week / month. This was helpful in framing proposals to spread the work load uniformly, regularize revenue distribution, improve customer satisfaction and roll-out new services.

Volume					Revenue				
Mean	Median	Mode	Min	Max	Mean	Median	Mode	Min	Max
120	115	101	82	171	1685	1610	1670	1200	2410

Figure – 7 :- Statistical Summary of Volume and Revenue Data

The shop currently uses both coal powered and gas powered iron-boxes. The iron-box is the principle equipment used to render laundry service. A study was done on the Advantage and Disadvantages of using each type of iron-box. Maintenance cost, Process involved and Lead times was captured in an excel sheet. This data provided inputs to analysis like **Process Re-engineering** and **BEP calculation**.

In order to gain insights into the cost incurred in running the shop I prepared a separate Cost sheet to list all the cost centers involved.

S.No.	Type of Iron-Box	Cost Of Purchase	Lifetime (Years)	Additional Cost	Depreciation per month
1	Coal	8000	1	100	766.6666667
2	Gas	12000	6	200	366.6666667

Figure – 8:- Coal-vs –Gas Iron box

Next, I calculated the Depreciation of Iron-box using **Straight Line Depreciation**, converting periodic expenses into monthly expenses and categorizing costs into Fixed / Variable to arrive at overall cost incurred on a monthly basis. **(Refer to Figure-9 Operational Cost- [LINK](#))**

S.No.	Cost Center	Fixed/Variable Cost	(Rs.)	(Rs.)
1	Rent	Fixed Cost	3,000.00	3,000.00
2(a)	Shop Maintanance	Fixed Cost	1,466.67	
2(b)	Mainatinance of Iron-box - Coal (Dust Net) = Rs. 200 - Gas (Gas Tube & Check-up) = Rs. 200	Fixed Cost	400.00	
2(c)	Depreciation - Gas Iron-box	Fixed Cost	366.67	
2(d)	Depreciation - Coal Iron-box	Fixed Cost	766.67	
2	Maintainence of Shop & Machniery		3,000.00	3,000.00
3(a)	Collection & Delivery Expenses	Fixed Cost	1,500.00	
3(b)	Daily Expenses	Fixed Cost	3,000.00	
3	Logistics & Operation Cost		4,500.00	4,500.00
4	Total Fixed Cost			10,500.00
5(a)	Cost of Gas	Variable Cost	2,600.00	
5(b)	Cost of Coal	Variable Cost	1,800.00	
5	Total Variable Cost		4,400.00	4,400.00
6	Total Cost			14,900.00

Figure – 9 :- Operational Cost Centers

This helped in identifying the **Breakeven Point**, the point where:-

- **Total Cost (Fixed + Variable) is equal to Total Revenue.**
- Assumption: - Sales Revenue, Variable Cost & Total Sales were taken as Average
- **Break Even Point (in Rs.)** = [Fixed Cost] / [Contribution Margin]
- **Contribution Margin (Per Unit)** = {[Revenue] - [Variable Cost]} / [Total Laundry]

The key outcome from Market Research was to introduce a new In-House service, Dry Cleaning. I researched the different models of Dry Cleaners available in the market, and narrowed down to a Brand and Model that will suit the current nascent stage of this business. Next, the credit options were collected and cost of financing this investment studied. Finally, information about the cost of running the machine, EMI charges and depreciation were calculated. The below Cost-sheet provides this break-up:-

S.No.	Cost Centers	(Rs.)
1	Cost Of Machine *	75,000.00
2	Financing Cost 12-Months (@15%)	11,250.00
3	Landing Cost of Machine	86,250.00
	Other Costs	
4	Running Cost - (Chemical, Power, etc.) Rs. per kg	50.00
5	Installment per month (24 Months) (Rs.)	3,593.75
6	Depreciation Per Month (5-Years)	1,250.00
*Source :- IFB Website - https://www.ifbappliances.com/rama-208?seeDetails=ref		

Figure-10:- Dry Cleaning Machine–Cost Centers **(Refer Figure-10 Dry Cleaning Machine- [LINK](#))**

This information was leveraged upon to extrapolate the Return-On-Investment (ROI), the time it will take for the investment to start generating profit and the conditions for it to do so.

3. Results and Findings

3.1. Break Even Point (B.E.P.)

(Refer to Figure-10 BEP(a)- [LINK](#))

B.E.P. shows how much the shop should earn per day to cover total cost incurred in running the business.

- **Break Even Point (in Rs.)** = [Fixed Cost] / [Contribution Margin]

S.No.	Component	Values (per Month)	Values (per Day)
1	Fixed Cost (Rs.) - Rent (Rs. 3,000) - Shop & Machinery (Rs. 3,000) - Logistics & Operation Cost (Rs. 4,500)	10,500.00	350.00
2	Average Variable Cost (Rs.) - Gas Cost (Rs. 2,600) - Coal Cost (Rs. 1,800)	4,400.00	146.67
3	Average Revenue / Sales (Rs.)	50,560.00	1,685.33
4	Total Laundry	3,600.00	120.00
5	Contribution Margin [(2-3)/4]	12.82	0.43
6	Break Even Point [1/5] (Laundry)	818.89	27.30
7	Average Revenue Per Laundry Unit (Rs.)	14.04	14.04
8	Break Even Point [6*7] (Rs.)	11,500.87	383.36

Figure – 10(a) :- BEP – Calculations

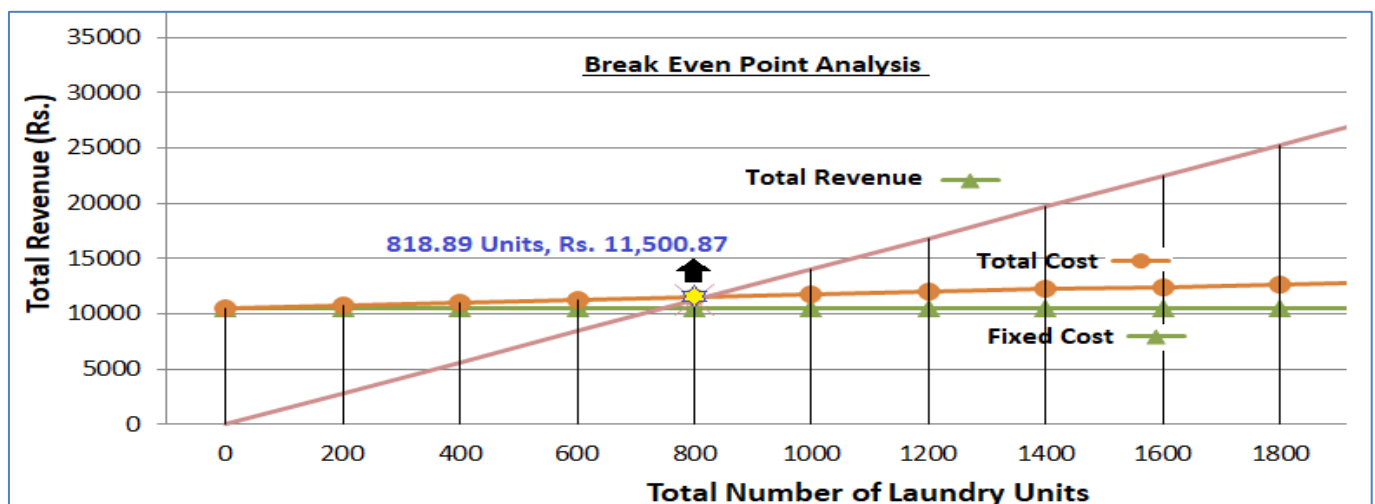


Figure – 10(b) :- BEP – Graphical Representation

Findings from the above analysis:-

- Results of Graphical representation of Fixed Cost, Variable Cost and Total Revenue is backed by the manual calculation of Break Even Point (BEP)
- The Shop should earn a Revenue of Rs.11,500.87 || Service 818.89 laundry units per/month **OR** Rs. 383.36 || 27.30 units per/day respectively
- BEP should be as low as possible, as anything above this point will start generating profit for shop
- The Fixed Cost Rs.10,500/- will always be incurred. The Variable Cost shows 2-factors Gas and Coal, which is approximately constant during the day
- Thus to reduce the BEP, it is crucial to spread this Variable Cost by efficiently using the operational time and servicing as much laundry as possible

3.2. Type of Iron-Box

(Refer to Figure-11 Iron Box- [LINK](#))

The shop currently uses both coal powered and gas powered iron-boxes, which have different set-up and operational capabilities.

Lead Time

S. No.	Type	Set-up time (Min)	Frequency per Day	Total Set-up time (min)	Comments
1	Coal	30	3	90	Loading-Unloading-Loading coal into the Iron box. Done 3-times each day.
2	Gas	5	3	15	Gas Iron box takes 5 minutes to heat-up and temperature can be controlled.

Figure – 11(a) :- Coal-vs-Gas Iron box – Lead Time

Coal Iron box Revenue Loss calculation is shown below:-

Description	Coal Iron-Box
Set-up Time (min)	30
Frequency Per Day	3
Set-up Time (min)	90
Avg. Laundry per min.	0.1667
Laundry During Set-up Time	15
Avg. Revenue per min. (Rs.)	2.34
Revenue During Set-up Time (Rs.)	210.67

Coal Iron-box consumes almost 90 minutes of operational time, which if put to productive use will generate approximately Rs.210.67. This productivity used will cover more than half the BEP value, thus increasing profits.

Figure – 11(b) :- Coal-vs-Gas Iron box – Revenue Lost

Summary of the comparison between the 2-types of Iron-boxes is presented below.

Clearly there are more advantages to only using a Gas Iron-box.

Advantages & Disadvantages (Coal/Gas Powered Iron-Boxes)

Characteristics	Coal Powered	Gas Powered
Temperature Control	Cannot Control Temperature	Ability to Control Temperature Control
Flexibility	Prioritize clothes based on Temperature need. Cotton need to be Ironed when temperature is high.	Can Iron clothes based on temperature needs, as temperature can be controlled.
Life Span	High corrosion because of coal, need to replace	Lesser wear and Tear, lasting atleast 6-years.
Servicing Capacity	Can service laundry based on Coal Temperature, which reduces with time.	Can service more laundry, as temperature can be increased or decreased as desired.
Lead Time	Loading-Unloading coal from Iron-box takes atleast 90-min each day. Reducing Operational time.	Atmost 15-mins spent each day to bring Iron-box to Operational mode. Increasing number of laundr serviced.
Health Issues	Coal exposure is hazardous to health, causes	No noteable health hazard
Climatic Condition	Difficult to set-up during raining days	No dependency on climatic condition
Weight	Iron-box is Heavy :- Increased - Infatigue &	Iron-box is Lighter :- Lower - Fatigue & Downtime for rest

Figure – 11(c) :- Coal-vs-Gas Iron box – Advantages & Disadvantages

3.3. Peak Volume and Revenue Analysis

(Refer to Figure-12 Peak Volume Revenue- [LINK](#))

Analyzing the data collected, it can be seen that volume of service offered by the shop varied throughout the week. Thus it was imperative to find Peaks and Trough in the demand for service. Thus **Volume of Laundry** and **Revenue** generated by the shop was plotted to represent distribution during the month.

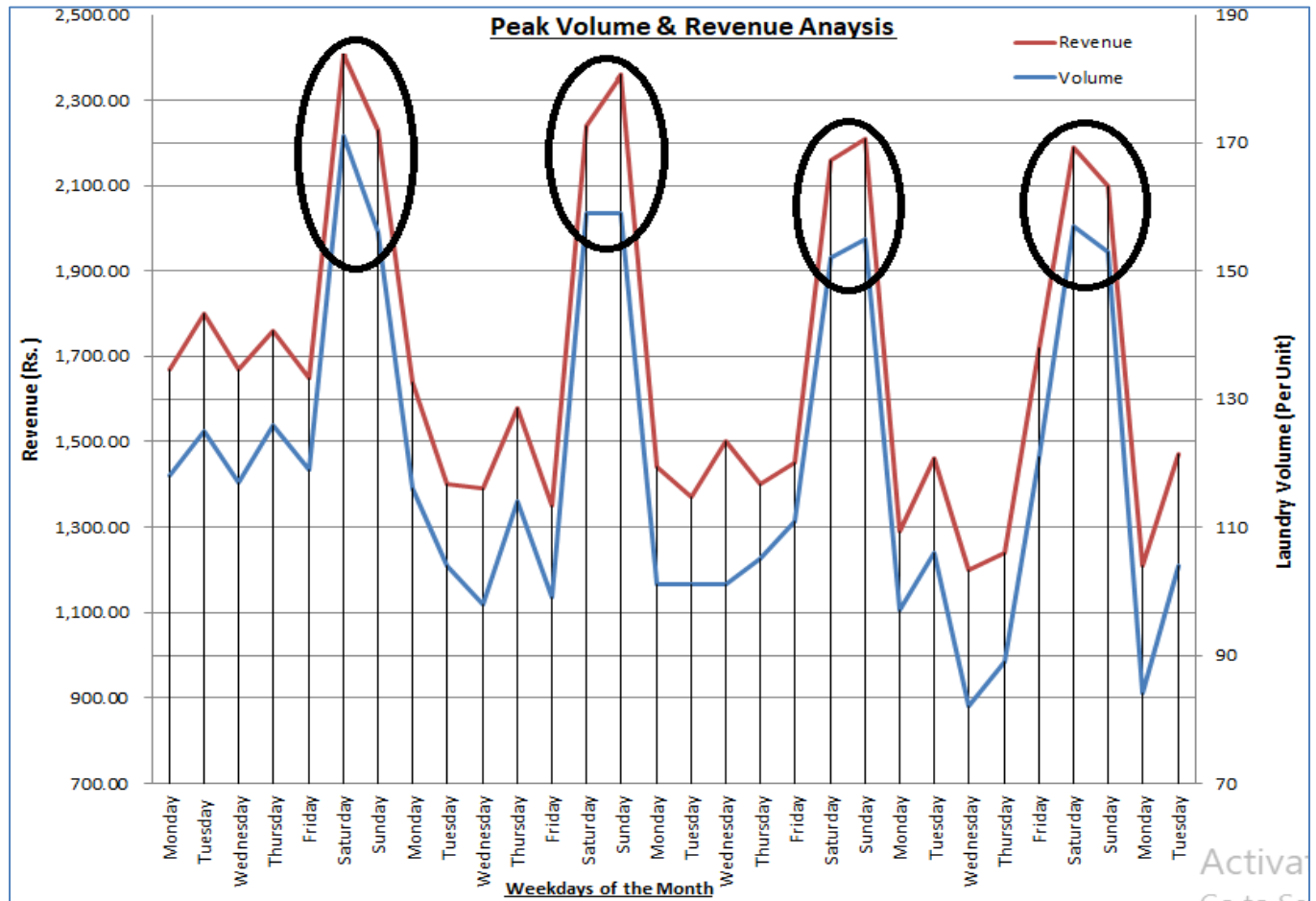


Figure – 12:- Graph of Volume & Revenue Distribution over a month

Summarizing the Results and Findings:-

1. Volume and Revenue **Peak during Weekend (Sat/Sun)** and **Trough during Weekdays**
2. Revenue varies between Rs.1,200 – Rs.2400. Thus, **BEP of Rs.383.36 is covered on all days.**
3. **Peaks and Trough** are a major **Bottleneck** faced by the shop, resulting in :-
 - i. Employees stretched during weekends
 - ii. Rejecting new orders during weekends
 - iii. Business Lost - Customers seek other option
 - iv. Commitment not honored, spill to Monday
 - v. Reputation and Customer satisfaction decreases
 - vi. Low Capacity Utilization during weekdays
4. Stagnation in business growth, as irregular demand and income affects expansion plans
5. Scope to implement new business strategies to Normalize Demand, bring new business, Increase Customer loyalty, Expand service offered which in-turn will increase the Revenue.

3.4. Process Re-engineering

(Refer to Figure-13 Process Re-eng- [LINK](#))

The current process stages and operational time spent at each stage is presented below. The time for Ironing Laundry was back populated:-

Proc ess	AS-IS Process	Duration (Min)	Frequ ency	Total Duration (Min)	Cumulative (Min)	Comments
1	Shop Opening	15	1	15	15	Mandatory - cannot be avoided
2	Laundry Collection & Sorting	15	4	60	75	Collection & Sorting can be clubbed along with Delivery
3	Setting-Up Coal Iron-Box	30	3	90	165	Instead of Using Coal it is proposed to use Gas Iron-box
4	Ancillary Activity	15	2	30	195	Required for Operation
5	Ironing Laundry	420	1	420	615	Productive Time - (More the better)
6	Ancillary Activity	15	2	30	645	Required for Operation
7	Laundry Sorting & Delivery	15	4	60	705	Collection & Sorting can be clubbed along with Delivery
8	Shop Closing	15	1	15	720	Mandatory - cannot be avoided
	Total Available Operation Time				720	

Figure – 13(a):- AS-IS Productivity Time Calculation Results (Ironing Laundry)

Findings from the current AS-IS process:-

- Shop opening / closing and ancillary activity are mandatory
- Almost 90-minutes is lost in setting-up coal powered Iron box daily
- Adhoc collection, sorting and delivery consumes 60+60 minutes
- Only 420-min (<60% Operational Time) is spent Ironing laundry

Based on the above findings, processes were Reorganized and Streamlined:- (TO-BE Process)

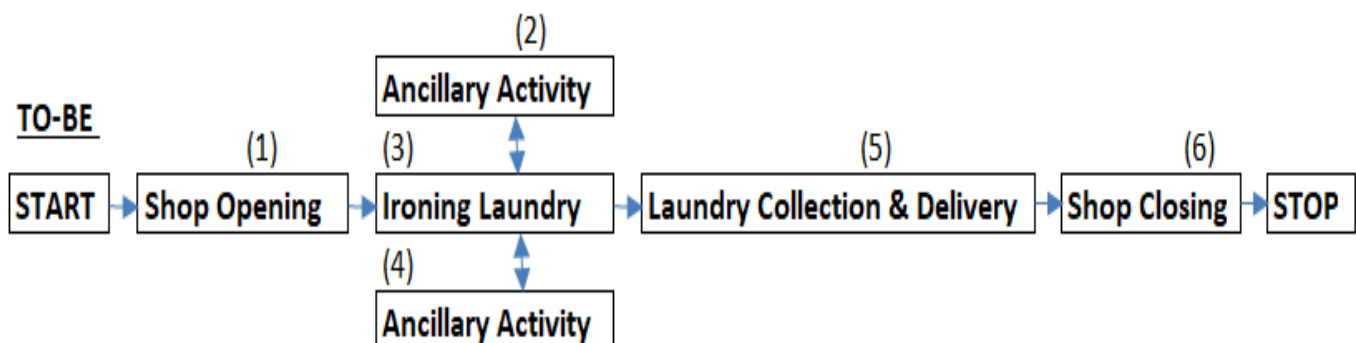


Figure – 13(b):- TO-BE Process Flow

Duration of time spent on each process was mapped to find the Real Time spent Ironing laundry:-

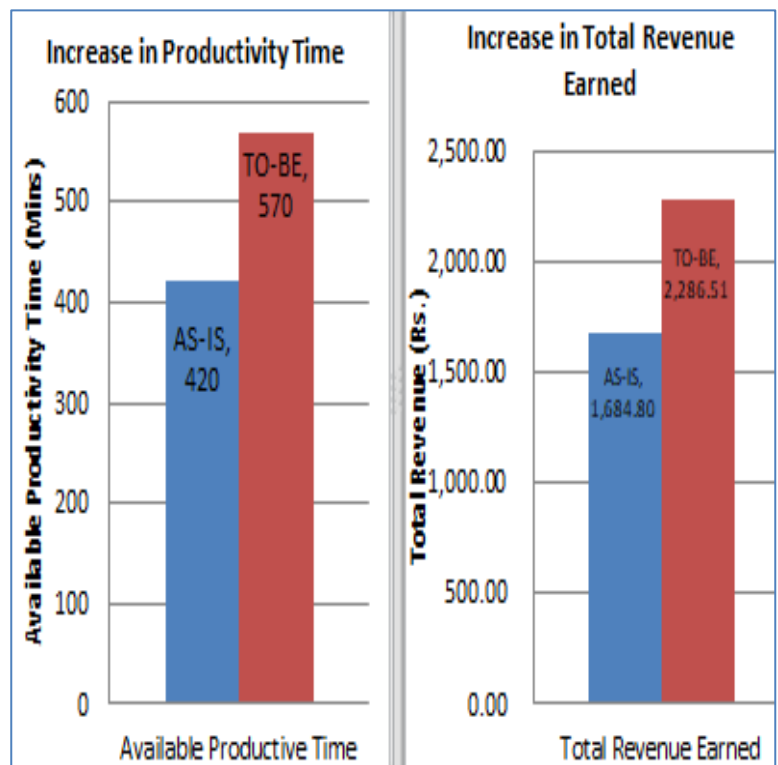
Proc ess	TO-BE Process	Duration (Min)	Frequ ency	Total Duration (Min)	Cumulative (Min)	Comments
1	Shop Opening	15	1	15	15	Mandatory - cannot be avoided
2	Ancillary Activity	15	2	30	45	Required for Operation
3	Ironing Laundry	570	1	570	615	Productive Time - Increased (More the better)
4	Ancillary Activity	15	2	30	645	Required for Operation
5	Laundry Collection, Sorting & Delivery	60	1	60	705	Collection and Sorting done EOD along with Delivery.
6	Shop Closing	15	1	15	720	Mandatory - cannot be avoided
	Total Available Operation Time				720	

Figure – 13(c):- TO-BE Productivity Time Calculation Results (Ironing Laundry)

Applying operational metrics to To-BE process and comparing results: -

	AS-IS	TO-BE
Available Productive Time	420	570
Laundry Served	120.00	162.86
Real Time Per Laundry	3.50	3.50
Revenue Per Laundry	14.04	14.04
Total Revenue Earned	1,684.80	2,286.51
Projected (%) Revenue Increase	-	35.71%
*Real Time = Taking into account time for other Support Activity		

Figure–13(d):- AS-IS vs TO-BE Comparison Calculation



Figure–13(e) AS-IS vs TO-BE Graphical Productivity and Revenue Comparison

Finding the change To-Be process brings to Breakeven Point:-

S.No.	Component	AS-IS (per Month)	TO-BE (per Month)
1	Fixed Cost (Rs.) - Rent (Rs. 3,000) - Shop & Machinery (Rs. 3,000) - Logistics & Operation Cost (Rs. 4,500)	10,500.00	10,500.00
2	Average Variable Cost (Rs.) - Gas Cost (Rs. 2,600) - Coal Cost (Rs. 1,800)	4,400.00	4,400.00
3	Average Revenue / Sales (Rs.)	50,560.00	68,595.30
4	Total Laundry	3,600.00	4,885.80
5	Contribution Margin [(2-3)/4]	12.82	13.14
6	Break Even Point [1/5] (Laundry	818.89	799.14
7	Average Revenue Per Laundry Unit (Rs.	14.04	14.04
8	Break Even Point [6*7] (Rs.)	11,500.87	11,219.68

Figure – 13(f):- AS-IS vs TO-BE Calculation of BEP

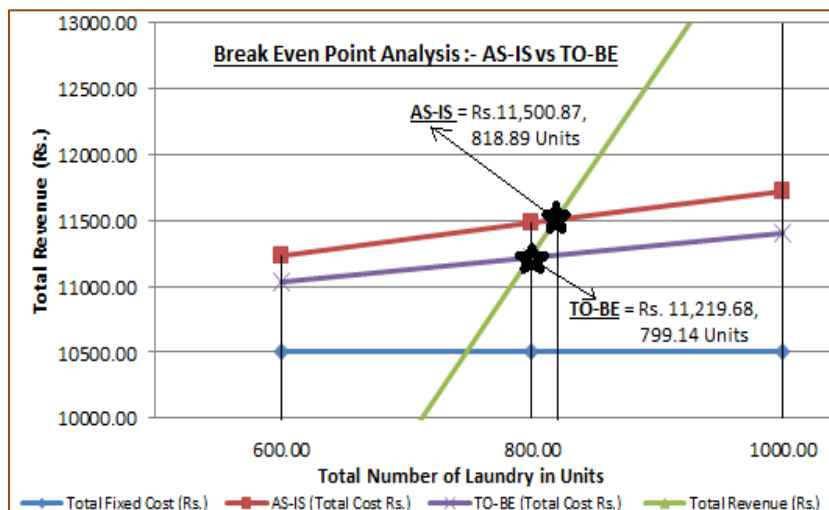


Figure – 13(g):- AS-IS vs TO-BE Graphical Representation of BEP

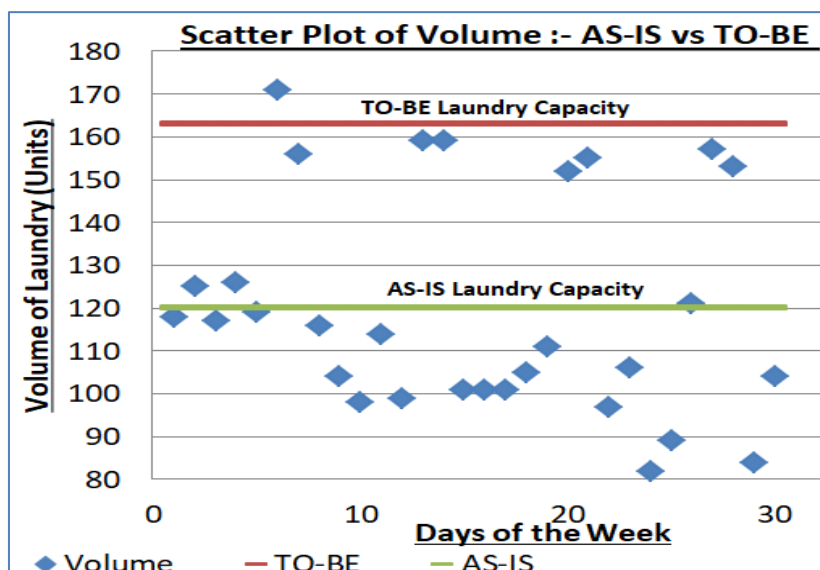


Figure – 13(h):- AS-IS vs TO-BE Laundry Volume Distribution

Summary of Results and Findings :-

- Using gas Iron Box, eliminates set-up time required for coal Iron-Box
- Laundry Collection and Sorting, can be done End-of-Day along with Delivery:- This Eliminates adhoc movement during business hours
- Revenue generated has increased by **35.71%**, more than 1/3 Increase
- The Break Even Point Reduces :-**
 - AS-IS = Rs. 11,500.87** || Service 818.89 laundry units per/month
 - TO-BE = Rs. 11, 219.68** || Service 799.14 laundry units per/month
- Current AS-IS process can service **120 Laundry Units Per Day**
 - From Scatterplot, there are **11-Days** where Demand is more than Capacity
- The proposed TO-BE process now can service **162.86 Laundry Units per Day**
 - From Scatterplot, there is only **1-day** where **Demand > Capacity**
- Proposed TO-BE Process will be able to service laundry more than 90% of the time
- Proposed TO-BE Process will enable taking up additional service request, which would have been avoided during the weekend when demand will be at peak

3.5 Volume Distribution and Return per Minute (RPM)

(Refer to Figure-14 RPM- [LINK](#))

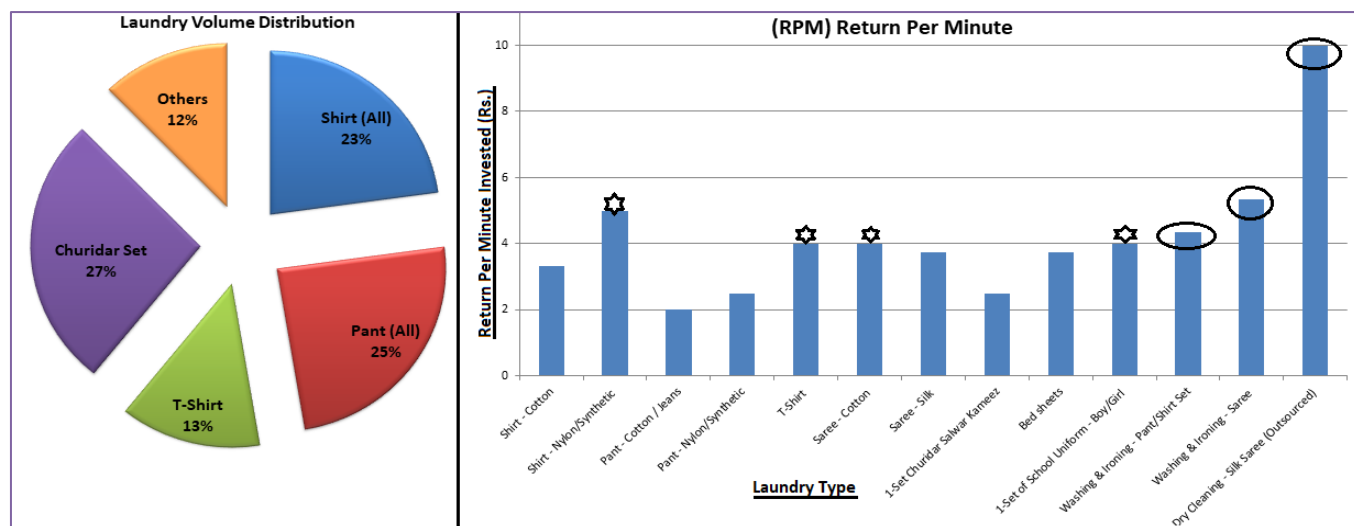


Figure – 14(a):- Laundry Distribution & Return Per Minute

Analysing the volume distribution and the Return Per Minute:-

- Laundry Volume Distribution shows that 4-categories (highlighted with STAR) namely Shirts, Pant, T-Shirt and Churidar contribute more than 80% of laundry serviced.
- Some laundry services (highlighted with a circle) provide greater returns than others, though they are provided as a package or need to be serviced outside the shop.

3.6 New Investment – In-House Dry Cleaning

(Refer to Figure-15 Dry Cleaning- [LINK](#))

The question that comes to everyone's mind when we speak of a new investment is 'Return on Investment' (ROI), time it takes for invested capital to start generating profit. Shorter the time duration the better. Below table provides Break-up of attributes like Operational Parameters and Estimations, Total Cost per month, Revenue Generated, Machinery Cost/Investment, Nominal Profit in each scenario.

S.No.	Break-up	1-Cycle/Day	2-Cycle/Day	3-Cycle/Day	4-Cycle/Day	5-Cycle/Day	6-Cycle/Day
1	Load capacity Per Cycle (kg)	8	8	8	8	8	8
2	Real World Load Per Cycle -75% (kg)	6	6	6	6	6	6
3	Number of Cycles per day	1	2	3	4	5	6
4	Real World Capacity (kg)	6	12	18	24	30	36
5	Average weight of per Laundry set (kg) (Note-1)	1.5	1.5	1.5	1.5	1.5	1.5
6	Number of Laundry Sets Serviced (Units)	4	8	12	16	20	24
7	Revenue Generated Per Day (Rs. 500) (Rs.)	1,200.00	2,400.00	3,600.00	4,800.00	6,000.00	7,200.00
8	Revenue Generated Per Month (30-Days) (Rs.)	36,000.00	72,000.00	1,08,000.00	1,44,000.00	1,80,000.00	2,16,000.00
9	Running Cost per kg (Rs. 50) - Per Month	9,000.00	18,000.00	27,000.00	36,000.00	45,000.00	54,000.00
10	Installment - EMI (Rs.) - Per Month	3,593.75	3,593.75	3,593.75	3,593.75	3,593.75	3,593.75
11	Depreciation Per Month (for 6-Years)	1,250.00	1,250.00	1,250.00	1,250.00	1,250.00	1,250.00
12	Total Cost Per Month	13,843.75	22,843.75	31,843.75	40,843.75	49,843.75	58,843.75
13	Nominal Profit Per Month (Rs.)	22,156.25	49,156.25	76,156.25	1,03,156.25	1,30,156.25	1,57,156.25
14	Landing Cost of Machine	86,250.00	86,250.00	86,250.00	86,250.00	86,250.00	86,250.00
15	Return on Investment (ROI) Period (in Months)	3.9	1.8	1.1	0.8	0.7	0.5

Figure – 15(a):- Dry Cleaning ROI calculation Results

Key Findings from this analysis:-

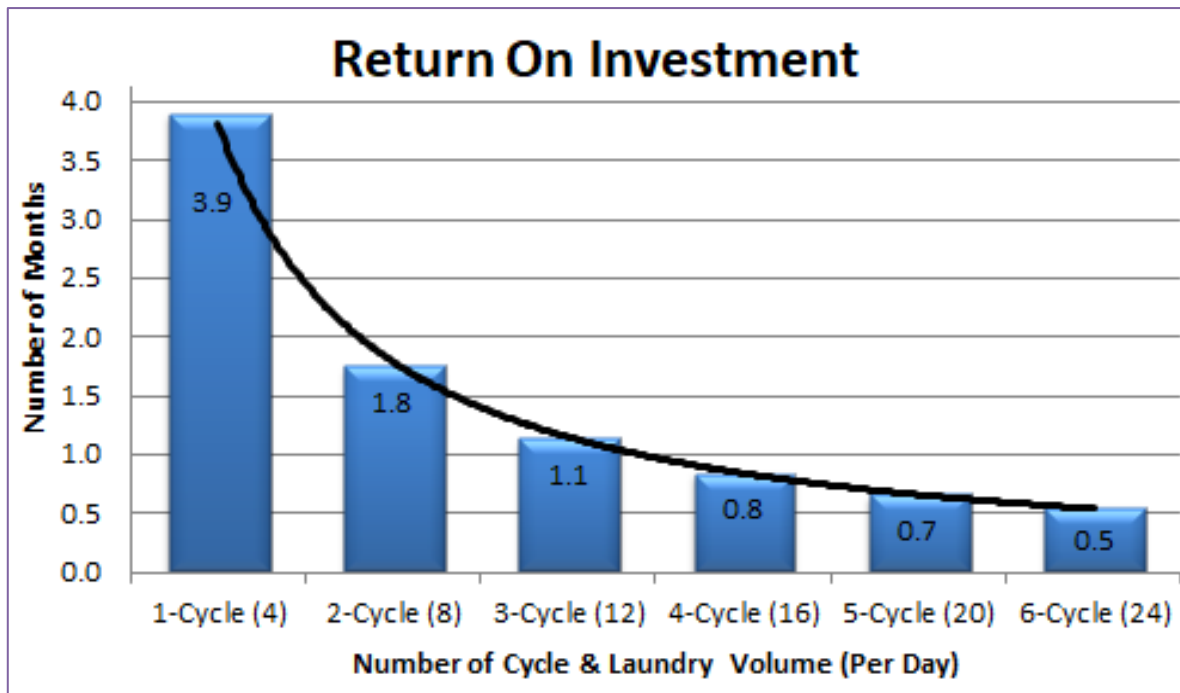


Figure – 15(b):- Dry Cleaning ROI calculation Results

- A. The Total Investment required to provide this service is Rs. 86,250. This includes cost of Financing.
- B. Taking a conservative approach of **75% efficiency** and low **capacity machine (8kg)** and machine run **1-cycle a day**, **ROI can be achieved in less than 4-months** (3.9-months)
- C. The **ROI period decreases** with increase in the number of cycles/volume serviced.
- D. This ROI can be achieved with a rate (Rs.300/-), lower than rate currently charged (Rs. 500/-)
- E. Though current volume of Dry Cleaning laundry is 15/month, it will ramp-up quickly :-
 - a. From interaction with customers, they used to go to a Dry Cleaning shop situated 2-km away. If this service is provided In-House and publicized Customers Volume will increase
 - b. When interacted with the Manager of a 4-Star Hotel nearby, they were very happy, as it will reduce their waiting Turn-Around-Time (TAT) and logistical issues travelling far for Dry Cleaning daily. They were ready to offer their support by giving their laundry for Dry Cleaning to the shop
 - c. Current charge is Rs.500/-. It will become attractive if reduced to Rs.300/-
 - d. Possibility of getting more orders from other laundry shops in the locality

4. Interpretation of Results and Recommendation

Proposal – 1:- **Use Only Gas Powered Iron-box**

Recommendation:- **Replace Coal Powered Iron-box with Gas Powered Iron-box**

The analysis of Coal & Gas Iron-box reveals that, there are a lot of tangible and intangible benefits in using only a Gas Powered Iron-Box.

Advantages of Using Gas Iron-Box:-

- a) **Health Benefit:-** Avoids infestation with Tuberculosis due to prolonged exposure to coal and Reduced Fatigue due to lighter weight of Gas Iron-box
- b) **Eliminates Lead-time:-** Using Gas Iron-box saves time on frequent Loading-Unloading of Coal
- c) **Flexibility:-** Allows Ironing irrespective of material, climate and temperature needs
- d) **Improves Efficiency and Productivity:-** Time and effort spent in Loading and Unloading of Coal can be used productively
- e) **Summing-up:-** Capacity to Service Increases, Breakeven Point Decreases, Revenue Increases and Employee Stress Decreases

Proposal – 2:- **New Business and Marketing Strategies**

Recommendation:- **Introduce Subscription Plans and Tariff slabs**

The S.W.O.T. analysis points out that the biggest strength the shop has is the strong reputation it maintains with customers in the locality. Results of interviews, surveys and market research points to the fact that there is potential to increase customer loyalty and garner new customers. One way to accomplish this is by launching new service subscription plans and introducing new pricing system.

Subscription Plans:-

- a) **Most customers demand service during the weekend:-**
The Subscription plan will aim to regularize the laundry that customer service during the week instead of taking them over the weekends.
- b) **Develop Customer Base with Commercial Establishments:-**
Have a tie-up plan with Commercial Establishments like the Blu IRIS hotel that is very close to the shop, to service their laundry needs. This will provide a steady stream of demand for the shop. Similar Subscription plans can also be extended to Paying Guest Hostels that function nearby.
- c) **Encourage Customers to avail more services:-**
The Subscription plan will attract customers to increase their laundry service needs. It will increase the volume of existing services, as well as clubbing them with other services like Washing + Ironing and Dry-Cleaning. This will not only increase the demand for services, but also promote services which are more profitable, higher Return Per Minute (RPM), thus increasing overall revenue generation.

Subscription Plan prepared with the shop Owner:-

(Refer to Figure-16 Subscription plan- [LINK](#))

Monthly Subscription Plan							
S. N	Plan Name	Frequency	Service Offering (for each visit)	Price (Rs.)	Volume	Estimated Actual Price	Percent Saving (%)
1	Ironing Package - 1	Once a Week (any day selected)	Ironing 20-Garments	1,000.00	80	1,120.00	11%
2	Ironing Package - 2	3-Times a Week (Mon, Wed, Fri)	Ironing 10-Garments	1,500.00	120	1,680.00	11%
3	Ironing Package - 3	2-Times a Week (Tue, Thu)	Ironing 15-Garments	1,500.00	120	1,680.00	11%
4	Laundry - Dry Cleaning (1)	Once a Week (any day selected)	Dry Clean 2-Garments	3,000.00	8	4,000.00	25%
5	Laundry - Dry Cleaning (2)	Once a Week (any day selected)	Dry Clean 5-Garments	8,000.00	20	10,000.00	20%
6	Laundry - Washing+Ironing (1)	3-Times a Week (Mon, Wed, Fri)	Ironing 10-Garments	20,000.00	120	24,000.00	17%
7	Laundry - Washing+Ironing (2)	2-Times a Week (Tue, Thu)	Ironing 10-Garments	14,000.00	80	16,000.00	13%
8	Commercial Laundry - Dry Cleaning (1)	Daily	Dry Clean 15-Garments	1,50,000.00	450	3,00,000.00	50%
9	Commercial Laundry - Dry Cleaning (2)	Daily	Dry Clean 10-Garments	1,00,000.00	300	1,50,000.00	33%
10	Commercial Laundry - Washing+Ironing (1)	Daily	Ironing 15-Garments	50,000.00	450	67,500.00	26%
11	Commercial Laundry - Washing+Ironing (2)	Daily	Ironing 20-Garments	60,000.00	600	90,000.00	33%

Figure – 16(a):- Subscription Plan Sample Template

Note: - The above plan can be kept flexible, so that they negotiated and customized to suit customer.

This not only will increase existing customer loyalty and get new customers, but will also help spread the demand throughout the week reducing weekend peak.

Pricing Slab/Discount

Pricing Slab/Discount		
S.No.	Laundry Volume	Discount
1	<=15 Units	NIL
2	16-25 Units	5%
3	26-50 Units	10%
4	51-100 Units	15%
5	>= 101 Units	20%

Pricing Slab will encourage customers to provide more laundry for service and also ensure there is less adhoc request for laundry service. This will reduce service interruption, as time spent responding to frequent service request can be avoided. Also, increase in demand for service will in turn increase revenue and profitability of the shop.

Figure – 16(b):- Sample Pricing Slab

Proposal – 3:-

Reorganize Business Operation Procedure & Practice

Recommendation: -

Streamline Operations and Introduce New Practices

Inefficient Procedures and Practices leads to loss of Productivity which in-turn affects the profitability of a business. Based on our analysis there are areas in the AS-IS process which can be altered and new practices introduced to improve efficiency. Removing redundant processes (Adhoc collection and Delivery), Upgrading to **New Practices (Gas Iron Box)** and **Reorganizing Activities (Clubbing delivery & collection)** will help increase productive time spent on core business activities This in-turn will result in increased revenue and profitability.

The Return Per Minute (RPM) analysis shows that the return from Dry Cleaning service provides the highest return for the time spent servicing, Analysis of the current volume distribution shows that there is a demand for Dry Cleaning among existing customers. This point is strengthened from the output of S.W.O.T. analysis and Market Research, which indicate that there is an untapped market for Dry Cleaning from Commercial Establishments like Paying Guest Hostels and Blu IRIS Hotel. The Dry Cleaning service is currently out-sourced which reduces the profit margin and capacity that can be handled regularly.

Recommendation: - Start In-House Dry Cleaning Service

This will require purchasing a Dry-Cleaning machine, as details below :-



Advantages of having an In-House Dry Cleaning faculty will help:-

1. Increase the variety of services offered by the shop
2. Reduce the Turn-Around-Time (TAT) for servicing
3. Increase Revenue generated and Profit Margin
4. Increase Customer Base as new customers will be attracted to the shop

Though current volume is 15/month, it will ramp-up:- (Reason's)

1. Currently customers generally use a Dry Cleaning shop 2-km away. If service is provided In-House and publicized, Volume will increase
2. A 4-Star Hotel nearby was very happy, as their waiting Turn-Around-Time (TAT) and logistical issues will reduce. They were ready to offer their support by giving their laundry for Dry Cleaning to this shop
3. Current rate is Rs.500/-. It will become attractive if reduced to Rs.300/-
4. High Probability of orders from other Laundry Shop's in the locality

Figure – 17:- Dry Cleaning Machine Model

Taking a very conservative number of servicing only 4-units per day, ROI will be **3.9 months**. Thus, making the investment in this new business service very attractive, that will turn profitable quickly. Further, this service can be ramped-up easily by purchasing a higher capacity Dry Cleaning machine.

5. Summary

“Change is the Only Constant”. The fundamental problem with the shop is that it was not able to adapt to changing times and customer behaviour. Thus it business was stagnant and found it difficult to expand. I firmly believe that the proposals backed by strong statistical data will enable the shop to Increase its Revenue and Profitability, and take the next leap to expand and grow its business.

******* THANK YOU *******